

Pacing the Frontier

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Key takeaways

Pacing isn't pausing: Calls to "pace the frontier" mean slowing the release of powerful artificial intelligence (AI) models to test and control them better, not halting development or spending.

Spending shifts, it doesn't disappear: Capital may move away from ever-larger training runs and toward safety infrastructure-testing, red-teaming¹, interpretability, monitoring, cybersecurity. This still requires significant compute and engineering investment.

Safety costs favor scale: Because rigorous safety work is expensive, the largest labs are best positioned to absorb it, which could accelerate consolidation among frontier AI even as smaller competitors struggle to keep up.

Neither side can afford to stop first: The United States and China each have too much incentive to keep advancing while the other does. Washington has signaled it won't mandate a broad slowdown, leaving the industry to build its own safety framework instead.

The AI Debate Just Changed

In recent days, Dario Amodei, Chief Executive Officer (CEO) of Anthropic, argued that the industry needs to "pace the frontier," giving safety and control more time to catch up with rapidly improving AI capabilities.

Markets initially heard something simpler: AI development is going to slow, and AI infrastructure spending will slow with it.

After watching the response from the industry, the US government and China over the past several days, we think that interpretation is increasingly unlikely.

The emerging outcome looks less like a pause and more like a race with more guardrails. Frontier development continues. The United States and China continue competing. But substantially more resources are devoted to understanding, testing, securing and controlling increasingly powerful models.

For investors, that is a very different conclusion from "the AI buildout is ending."

The Problem Is Changing

For the last several years, the big question about AI has been simple:

Can the models keep getting better?

So far, the answer has largely been yes.

More compute, better models and better techniques have continued to produce more capable systems. Increasingly, AI models can write software, conduct research, use tools and operate with growing autonomy.

That progress has created a new question:

Can we understand and control these systems as quickly as we can make them more powerful?

Increasingly, the people building the models are telling us that it cannot.

That is the most important part of Amodei's argument. He is not arguing that AI has stopped working. In some respects, he is arguing the opposite. Capabilities are advancing quickly enough that alignment, interpretability, evaluation, cybersecurity and governance are struggling to keep up.

And the industry's response over the past several days is telling. Sam Altman, Elon Musk and Demis Hassabis have all expressed support for important elements of Amodei's proposal, while offering more incremental approaches to implementation. Altman supported deeper access for independent evaluators. Hassabis pointed toward common industry standards. Others are proposing variations on third-party evaluation and industry-led safety mechanisms.

There is plenty of disagreement about exactly how far to go and how much government should be involved. But there appears to be growing agreement on the underlying problem: safety and control need to catch up with capability.

More Safety Still Requires More Infrastructure

The first market reaction to "pacing" was understandable. If the labs slow down, perhaps they need fewer graph processing units (GPU), fewer data centers and less power.

Amodei's proposal does not envision simply turning off the training clusters. It calls for greater investment in alignment, interpretability, testing, evaluation, monitoring and security as capabilities advance.

Those activities require significant technical resources. That work is compute intensive:

Models need to be tested repeatedly across enormous numbers of interactions and undergo sophisticated red-teaming.

Researchers need to understand why models behave the way they do.

Increasingly autonomous systems require stronger cybersecurity and monitoring.

Problems found during testing may require additional post-training.

Agents must be evaluated inside increasingly complex environments before release.

That does not mean every dollar spent on alignment automatically becomes an additional dollar of AI capital spending. If frontier labs materially reduce the size or frequency of their largest training runs, some infrastructure demand could be deferred.

But we think there is a more useful way for investors to frame the question: every unit of frontier progress may now require more infrastructure around it.

We may need more evaluation, more post-training, more inference, more security, more monitoring and more experimentation for each step forward in capability.

That could make the next phase of the AI infrastructure buildout more complicated and more safety-intensive without necessarily making it smaller.

A Self-Regulatory Path Is Emerging

The second important development has been the response from Washington. One of the risks coming into the weekend was that the industry's own safety concerns could trigger a government-mandated slowdown in frontier development.

For now, that appears less likely.

President Trump has rejected calls for the United States to broadly slow AI development, arguing that doing so could weaken the country's position relative to China. He reiterated that view at the All-In Summit on Monday. Nvidia CEO Jensen Huang similarly argued that the United States can continue leading in AI while improving safety.

That changes the near-term policy calculus. Rather than Washington imposing a broad brake on frontier development, the emerging path may require the industry to build much of the safety framework itself.

We are beginning to see what that could look like:

A credible self-regulatory framework could provide a useful middle ground.

It could improve safety and create greater predictability for companies and investors without imposing broad limits on training or innovation. Clearer standards could also reduce the risk of a much more aggressive government response after a serious AI-related incident.

There are important caveats. Self-regulation could prove insufficient. Standards could vary between companies. The fixed costs of compliance could favor the largest labs. Congress, individual states or future administrations could ultimately take a more interventionist approach.

But from our perspective, there is an important distinction between building guardrails around innovation and restricting the innovation itself.

China Makes a True Pause Difficult

Then there is China. China's reaction highlights why a sustained global slowdown would be extraordinarily difficult.

Beijing pushed back sharply against Amodei's argument that the United States should protect its technological lead over China, criticizing what it described as "fearmongering" and confrontation.

But China is not saying that AI carries no risk. Chinese leaders have repeatedly discussed the need for AI governance, control and safety. At the same time, China continues pushing aggressively to improve its models, expand AI adoption and develop its domestic AI ecosystem. Those positions are not necessarily inconsistent. China understands the risks associated with increasingly powerful AI. But it also has enormous strategic and economic incentives to continue advancing the technology.

Why would China voluntarily slow while it is trying to close the gap with the United States? And why would American labs materially slow if they believe Chinese labs will continue advancing? This is the fundamental constraint on the pacing debate. Neither side has much incentive to stop if the other continues moving. That makes a sustained global pause unlikely.

The more plausible equilibrium: everyone keeps running, while spending more to make the race safer.

A Race with More Guardrails

We increasingly think this develops through four reinforcing stages.

More engineering and compute go toward alignment, interpretability, evaluation, cybersecurity and control. Independent third parties play a larger role in assessing frontier models. The bar for releasing increasingly autonomous systems rises. And safety becomes another important layer of the AI technology stack.

This also creates second-order effects.

The largest frontier labs may gain an advantage because they can absorb the substantial fixed costs of safety infrastructure and compliance. Smaller model developers may find those costs more difficult to bear.

Cybersecurity becomes more important as AI systems gain greater autonomy and access to tools and data.

Inference and evaluation workloads become more significant as companies spend more time testing what models can do before broadly deploying them.

And demand may increasingly move beyond the GPUs used to train models toward the broader infrastructure required to operate, test and secure them.

None of this suggests that the risk has disappeared.

A serious AI incident could change the regulatory environment very quickly. Governments may eventually decide that self-regulation is insufficient. The US policy approach could also change with a future administration.

But those are risks to monitor, not reasons to assume that today's safety debate necessarily ends the AI investment cycle.

The Question That Still Matters

For long-term investors, the fundamental question remains unchanged:

Does making AI more capable continue to create economic value?

If models stop improving, or if businesses fail to generate meaningful productivity gains from them, the infrastructure cycle will eventually slow. No amount of enthusiasm can overcome poor economics indefinitely.

But if capabilities continue improving and customers continue finding valuable uses for AI, it is difficult to imagine companies or countries collectively deciding that they have enough intelligence. Instead, we are likely to spend enormous amounts of capital trying to capture the benefits of increasingly powerful AI while reducing its risks.

That means the important change may be less about the direction of AI spending than about its composition and intensity.

The first phase of this investment cycle was dominated by the race to build more intelligence. The next phase looks more complicated.

We still need to build intelligence. But increasingly, we also need to understand it, test it, secure it and control it. All of that requires infrastructure. After watching the debate evolve over the past several days, we therefore come away somewhat more constructive than we were when it began.

The frontier is unlikely to stop moving. The race is unlikely to end. The first phase of the AI investment cycle was largely about building more intelligence. The next phase may increasingly be about something harder: building intelligence we can trust.

Endnote:

Red-teaming: a structured, authorized exercise where a group of security experts acts as an adversary to test an organization's defenses, plans, or systems.

WHAT ARE THE RISKS?

All investments involve risks, including possible loss of principal.

Equity securities are subject to price fluctuation and possible loss of principal.

Small- and mid-cap stocks involve greater risks and volatility than large-cap stocks.

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